Removing the extra smoothness unit for Métivier sub-Laplacians with drift

Candidate solution packet for arXiv:2605.02556

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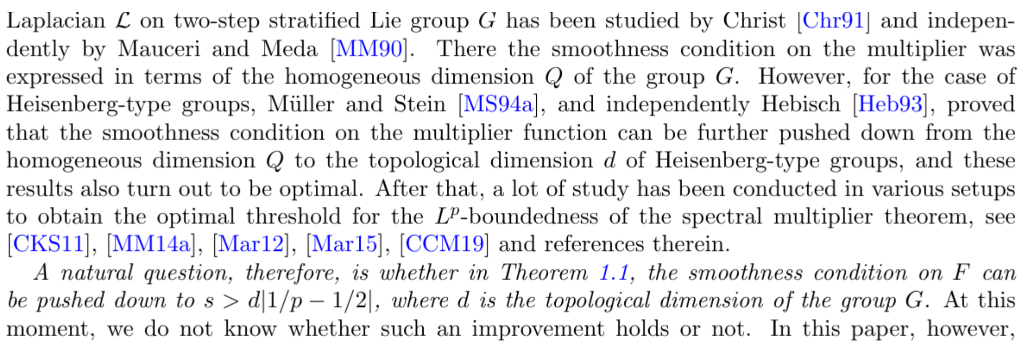
Outcome. Let G be a Métivier group of topological dimension d. The multiplier theorem with drift holds under

$$s > d|1/p - 1/2|,$$

removing the extra $+1$ in the source's condition $s > (d+1)|1/p - 1/2|$. The new observation is geometric: the positive character localizes a large propagation ball to a thin cap, whose polynomial volume is much smaller than that of a whole outer annulus.

1. The source question

Write the stratified Lie algebra as $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$, with $d_i = \dim \mathfrak{g}_i$, $d = d_1 + d_2$, and homogeneous dimension $Q = d_1 + 2d_2$. For a nontrivial positive character χ, let $\mathrm{d}\mu_X = \chi\ \mathrm{d}g$, $b_X = |X|/2$, and $\mathfrak{D}_X = (\mathcal{L}_X - b_X^2)^{1/2}$. Garg–Singh prove the multiplier theorem with the threshold $(d+1)|1/p-1/2|$ and ask for $d|1/p-1/2|$.


Laplacian $\mathcal{L}$ on two-step stratified Lie group G has been studied by Christ [Chr91] and independently by Mauceri and Meda [MM90]. There the smoothness condition on the multiplier was expressed in terms of the homogeneous dimension Q of the group G. However, for the case of Heisenberg-type groups, Müller and Stein [MS94a], and independently Hebisch [Heb93], proved that the smoothness condition on the multiplier function can be further pushed down from the homogeneous dimension Q to the topological dimension d of Heisenberg-type groups, and these results also turn out to be optimal. After that, a lot of study has been conducted in various setups to obtain the optimal threshold for the L^p-boundedness of the spectral multiplier theorem, see [CKS11], [MM14a], [Mar12], [Mar15], [CCM19] and references therein.

A natural question, therefore, is whether in Theorem 1.1, the smoothness condition on F can be pushed down to $s > d|1/p-1/2|$, where d is the topological dimension of the group G. At this moment, we do not know whether such an improvement holds or not. In this paper, however,

Figure 1: The exact question on source PDF page 3.

The notation $\mathscr{L}^{q,-1/q,\Theta}_{s,\infty}$ below is exactly the source's scale-invariant Sobolev class of even functions holomorphic in the strip $|\operatorname{Im} z| < \Theta$, with boundary trace measured at smoothness s.

Theorem 1 (Optimal-form Métivier theorem)**.** *Let G be a Métivier group and $X \neq 0$. Let $p \in [1,\infty] \setminus \{2\}$ and choose $q \in [2,\infty)$ by*

$$\frac{1}{q} = \frac{|2/p-1|}{2}, \qquad \Theta_{X,p} = |2/p-1| b_X.$$

If

$$F \in \mathscr{L}^{q,-1/q,\Theta_{X,p}}_{s,\infty} \quad \textit{for some} \quad s > d|1/p-1/2|,$$

then $F(\mathfrak{D}_X)$ is bounded on $L^p(\mu_X)$ when $1 < p < \infty$; for $p = 1$ it maps the local Hardy space $\mathfrak{h}^1(\mu_X)$ to $L^1(\mu_X)$, and for $p = \infty$ it maps $L^\infty(\mu_X)$ to $\mathfrak{bmo}(\mu_X)$.

Proof intuition. The source's local kernel estimates already require only $s > d/2$. Its global estimate uses finite propagation and then bounds a character-weighted ball by an entire homogeneous annulus. But $\chi(x,u) = e^{\langle a,x\rangle}$, $|a| = 2b_X$: after the exponential factor $e^{2b_X r}$ is removed, only a thin cap near $x = re_1$ contributes. The cap's central fibers shrink as the square root of the distance to the boundary. Keeping this shrinkage removes the extra summability loss.

2. The cap estimate

Let ρ be the Carnot–Carathéodory distance and $B_\rho(e,r)$ its ball.

Lemma 1 (Character-weighted cap). *For $r \geq 1$ and $0 \leq \beta < d_2/2$,*

$$\int_{B_\rho(e,r)} \frac{\chi(x,u)}{|x|^{2\beta}}\,\mathrm{d}x\,\mathrm{d}u \leq Ce^{2b_X r} r^{(d_1+3d_2-1)/2-2\beta}. \tag{1}$$

Proof. Every positive character kills the commutator layer, so $\chi(x,u) = e^{\langle a,x\rangle}$ with $|a| = 2b_X$. Rotate the first layer so $a = 2b_X e_1$ and write $x = te_1 + z$.

We first record a fiber bound. If $(x,u) \in B_\rho(e,r)$, choose a horizontal path $\gamma : [0,r] \to \mathfrak{g}_1$ of speed at most one, with $\gamma(0) = 0$ and $\gamma(r) = x$. Put $v = x/r$ and $q = \gamma' - v$. Since $\int_0^r q = 0$,

$$\int_0^r |q|^2\,\mathrm{d}t \leq r - \frac{|x|^2}{r} = \frac{r^2 - |x|^2}{r}.$$

The step-two endpoint formula is $u = \frac{1}{2}\int_0^r [\gamma(t), \gamma'(t)]\,\mathrm{d}t$. Subtracting the straight path and using the boundedness of the bracket gives

$$|u| \leq C\left(r\int_0^r |q|\,\mathrm{d}t + \left(\int_0^r |q|\,\mathrm{d}t\right)^2\right) \leq Cr\sqrt{r^2 - |x|^2}.$$

Thus, for fixed $|x| < r$, the allowed central fiber has volume at most

$$Cr^{d_2}(r^2 - |x|^2)^{d_2/2}. \tag{2}$$

The region $t \leq r/2$ contributes at most $Ce^{b_X r} r^{Q-2\beta}$; for $r \geq 1$ the spare exponential absorbs the polynomial difference and makes this no larger than the right side of (1). On the cap $r/2 < t < r$, we have $|x|^{-2\beta} \leq Cr^{-2\beta}$. Using (2), integrating first in $z \in \mathbb{R}^{d_1-1}$, and then setting $y = r - t$, gives

$$\begin{aligned} &Cr^{d_2-2\beta}\int_{r/2}^r e^{2b_X t}(r^2 - t^2)^{(d_1+d_2-1)/2}\,\mathrm{d}t \\ &\quad\leq Ce^{2b_X r} r^{d_2-2\beta+(d_1+d_2-1)/2}\int_0^\infty e^{-2b_X y} y^{(d-1)/2}\,\mathrm{d}y, \end{aligned}$$

which is exactly (1). ☐

3. From the cap to the multiplier theorem

The source proves, for every $0 \leq \beta < d_2/2$, the first-layer weighted Plancherel estimate

$$\big\||x|^\beta \mathcal{K}_{F(\sqrt{\mathcal{L}})}\big\|_2 \leq C\left(\int_0^\infty |F(\lambda)|^2 \lambda^{Q-2\beta}\frac{\mathrm{d}\lambda}{\lambda}\right)^{1/2}. \tag{3}$$

If $\operatorname{supp}\widehat{F} \subset [-r,r]$, finite propagation puts the convolution kernel in $B_\rho(e,r)$. Cauchy–Schwarz, Lemma 1, and (3) therefore yield

$$\|\chi^{1/2}\mathcal{K}_{F(\sqrt{\mathcal{L}})}\|_1 \leq Ce^{b_X r}(1+r)^\vartheta \|(1+|\cdot|)^a F\|_2, \tag{4}$$

where

$$\vartheta = \frac{d_1 + 3d_2 - 1}{4} - \beta, \qquad a = \frac{Q}{2} - \beta - \frac{1}{2}. \tag{5}$$

Now apply the conditional multiplier theorem of Martini–Ottazzi–Vallarino. With Fourier exponent 2, its required endpoint smoothness is any number strictly larger than

$$\max\{\sigma,\ \vartheta + 1,\ a + 1/2\}. \tag{6}$$

Here $\sigma > d/2$ is the local kernel threshold; the source has already proved the two required local estimates for every such σ. Letting $\beta \uparrow d_2/2$ in (5),

$$a + \frac{1}{2} \longrightarrow \frac{d}{2}, \qquad \vartheta + 1 \longrightarrow \frac{d+3}{4} \leq \frac{d}{2},$$

because a Métivier group has $d_1 > d_2 \geq 1$, hence $d \geq 3$. Thus the infimum of (6) is $d/2$. The 2019 theorem multiplies this endpoint by $2|1/p - 1/2|$, giving precisely the hypothesis in Theorem 1. ☐

Integer-derivative corollary. The same argument and the standard embedding used by the source replace its Corollary 1.3 condition by

$$F \in H^{\infty}(S_{\Theta_{X,p}}; N), \qquad N > d|1/p - 1/2|.$$

4. What is and is not resolved

The proof fully removes the extra unit for the Métivier class studied in the source. Lemma 1 is purely step-two geometry and in fact holds for every step-two stratified group. Consequently it also gives a conditional transfer principle: the topological threshold follows whenever the local $d/2$ kernel estimates and the first-layer weighted Plancherel bound (3) are available up to $\beta < d_2/2$.

The unrestricted question in the source is phrased for arbitrary two-step groups. In degenerate groups, the source's proof of (3) uses precisely the Métivier nondegeneracy that is absent; the corresponding sharp no-drift multiplier theorem is itself not known in this generality. No claim is made for those groups.

Upgrade audit. Six focused routes were checked: locating the loss; square-summing Fourier time pieces; testing a radial counterexample mechanism; proving the cap geometry; inserting its exponents into the abstract theorem; and attempting the arbitrary two-step upgrade. The cap route closes the Métivier case. The last upgrade stops at the missing weighted Plancherel estimate, not at a summability defect.

Novelty boundary. Bounded primary-source searches through 17 August 2026 found the source, the 2019 abstract theorem, and adjacent sharp no-drift results, but no later paper removing the $+1$ in the drift theorem. The argument should still be checked by an expert in subelliptic spectral multipliers before dissemination.

References

[1] N. Garg and J. Singh, *On Spectral multiplier theorem for sub-Laplacians with drift on Métivier groups*, arXiv:2605.02556 (2026).

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[3] L. Niedorf, *Spectral multipliers on Métivier groups*, arXiv:2412.07920.